

SHORT COMMUNICATION

Testing the Performance of Macroinvertebrate-Based Indices of Intermittence in Rivers Beyond Their Region of Development

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ABSTRACT

Intermittent rivers represent more than half of the length of the global river network, and their extent is predicted to increase. Although flow intermittence is a natural phenomenon, it is exacerbated by the combined and interacting effects of climate change and anthropogenic pressures. This increasing intermittence represents a major threat to freshwater biodiversity and highlights the pressing need for indices that can assess the responses of freshwater biota to drying events. However, most such indices remain untested beyond their region of origin, limiting their widespread use. We addressed this gap by testing the capacity of two macroinvertebrate-based indices that respond to flow intermittence (the drought effect of habitat loss on invertebrates [DEHLI] index and the monitoring intermittent streams index [MIS-index]), developed and evaluated in United Kingdom, to distinguish between perennial and intermittent river sites in Italy. Scores for both indices and index-derived richness metrics (i.e., the taxa richness of groups of environmental association classified by the indices) were higher at perennial than at intermittent sites. Among index-derived richness metrics, only those including rheophilic taxa were lower at intermittent sites, whereas others were comparable at perennial and intermittent sites. Our results suggest that both indices have the potential for use beyond their original region of development, with a better performance of the DEHLI index due to its family-level taxonomic resolution. However, their responsiveness could be enhanced by regional adaptation to increase the pool of scoring taxa. In addition, the development of conceptual frameworks predicting assemblage composition in intermittent rivers across different biogeographical regions could guide the future development of new indices for flow intermittence.

1 | Introduction

More than 50% of the global river network experiences flow intermittence, with dry periods spanning days to years (Messenger et al. 2021). While river drying is a natural phenomenon, the extent of intermittent rivers is increasing due to climate

change and growing human pressures (Döll and Schmied 2012; Mimeau, Künne, Branger et al. 2024a; Trambly et al. 2021), with previously perennial rivers becoming intermittent (Datry et al. 2023). This increasing river drying represents a major threat to freshwater biodiversity, reducing aquatic diversity primarily through the loss of sensitive taxa (Leigh and Datry 2017).

In this context, tools are needed to monitor how aquatic biota respond to intermittence, during seasonal dry phases as well as unpredictable drought events (Skoulikidis et al. 2017; Stubbington et al. 2017). As such, new indices have been developed to assess the responses of river biota, in particular aquatic macroinvertebrate communities, to changing habitat conditions (England et al. 2019; Straka et al. 2019). However, these indices have been developed and tested in specific environmental contexts (i.e., within individual countries or ecoregions), and their performance beyond their regions of origin remains largely untested (but see Theodoropoulos et al. 2021).

Two indices developed in England, United Kingdom, characterize macroinvertebrate community responses to hydrological variability, including stream drying: the drought effect of habitat loss on invertebrates index (DEHLI; Chadd et al. 2017) and the monitoring intermittent streams index (MIS-index; England et al. 2019). DEHLI quantifies community responses to changing in-channel habitat conditions during droughts by assigning drought intolerance scores (DIS) to macroinvertebrate families (or genera) within ecological groups associated with habitats lost at key stages of channel drying (i.e., *flow-decrease*, *loss-lateral*, *loss-longitudinal*, *pool-shrinking*, *pool-drying*, *total-dry*; Table S1). MIS-index characterizes ecological responses to habitat changes associated with flow intermittence by averaging weighted taxa richness (largely at genus or species level) in six habitat-association groups (i.e., MIS-groups: *lotic-fast*, *lotic*, *generalist*, *lentic*, *semi-aquatic*, *terrestrial*; Table S1). By characterizing community responses to changing hydrological conditions, these indices could also support the interpretation of biomonitoring data in river health assessments. However, the capacity of DEHLI and MIS-index to characterize community responses to drying has only been tested within United Kingdom lowland calcareous systems (i.e., the chalk streams; Bunting et al. 2020; England et al. 2019; White et al. 2019), whereas index performance in other contexts remains unexplored.

We addressed this gap by evaluating the ability of DEHLI and MIS-index to distinguish between macroinvertebrate communities collected in perennial and intermittent river sites in Italy, thus testing the transferability of indices across environmental contexts. Our central hypothesis was that indices developed in one region can be effectively applied in other environmental contexts if there is sufficient overlap in taxa pools. As such, we predicted that both DEHLI and MIS-index would distinguish between perennial and intermittent sites due to the overlapping regional taxa pools (i.e., 77%, 54%, and 26% at the family, genus and species levels, respectively; see Figure S1). We also predicted that DEHLI would outperform the MIS-index because it is based mainly on family-level data, rather than a mixed taxonomic level dominated by genera and species, and is thus more likely to share taxa with those in other ecoregions (e.g., Schmidt-Kloiber and Hering 2015).

2 | Methods

2.1 | Test Dataset: Study Area and Design

The performance of DEHLI and MIS-index was tested using a dataset collected from 22 perennial (P) and 18 intermittent (I) sites across five streams in the Po River basin (Emilia-Romagna

region, northern Italy) in May–June 2017 (Figure 1). The climate in the study area is temperate/cold (Köppen–Geiger classification Cfa, Cfb, and Dfb; Beck et al. 2018), with mean annual temperature ranging from 5°C to 14°C and annual precipitation from 600 to 2300 mm (both calculated for the period 1961–2010 in Emilia-Romagna region) (Antolini et al. 2016). Site elevation varied between 174 and 820 m a.s.l. (mean $\pm$ SD, 469 $\pm$ 162 m). Land use (calculated for each sub-catchment from CORINE land cover data 2018) is mainly forests or semi-natural (mean $\pm$ SD, 73% $\pm$ 21%, according to CORINE land cover nomenclature).

Rivers and streams in the study area are both surface water and groundwater-fed, with high discharge in early spring and autumn and low discharge in summer. Each site's flow permanence regime was classified as P or I based on regulatory authority discharge data (when available, <https://www.arpae.it>), aerial photos from Google Earth, and information provided by local residents. Classifications were then validated using field observations and hourly water temperature data logged from late July to mid-November (Onset HOB0). We classified I-sites as those that dried for ≥ 2 consecutive days in this period (Blasch et al. 2004). The number of dry days for the 18 I-sites varied between three and 70 days (mean $\pm$ SD, 53 $\pm$ 21 days), with flow resuming at I-sites in early November. P-sites were classified as those with continuous flow.

Macroinvertebrates were collected from mid-May to mid-June 2017. A Surber net (0.05 m² frame area, 500- μ m mesh) was used to collect six macroinvertebrate samples per site, encompassing all available habitats to maximize the number of taxa collected. One site had only isolated ponds at the time of sampling; consequently, only two samples were collected. Each macroinvertebrate Surber sample (236 in total) was preserved with 90% ethanol and stored until processing in the laboratory. All material was processed and specimens identified to family or genus level (Campaoli et al. 1994; Tachet et al. 2010), their life stage (i.e., juvenile, pupa or adult) noted, and each counted.

2.2 | Statistical Analysis

We assigned each taxon a DIS following Chadd et al. (2017) and to a MIS-group following England et al. (2019) (Table S1). Taxa without a DIS and/or a MIS-group were excluded from the dataset (see Section 4 and Table S2). The DEHLI index was then calculated as the sum of the DIS values divided by the number of scoring taxa (Chadd et al. 2017). The MIS-index was calculated as the weighted mean (i.e., the sum divided by the total number of scoring taxa in a sample) of the taxa in each MIS-group multiplied by its weighting factor (England et al. 2019). For the calculation of both indices, we aggregated taxa at the taxonomic level required by each index (i.e., mainly family level for DEHLI and a mixed level for MIS-index; Table S2). We also calculated the aggregated taxa richness in each DEHLI group (six metrics) and each MIS-group (six metrics). Metrics were not calculated for the *total-dry* DEHLI group or the *terrestrial* MIS-group because 0–1 taxa were recorded for these groups.

We compared index scores and richness metric values at P and I sites using linear mixed-effect models (LMM, for indices) and

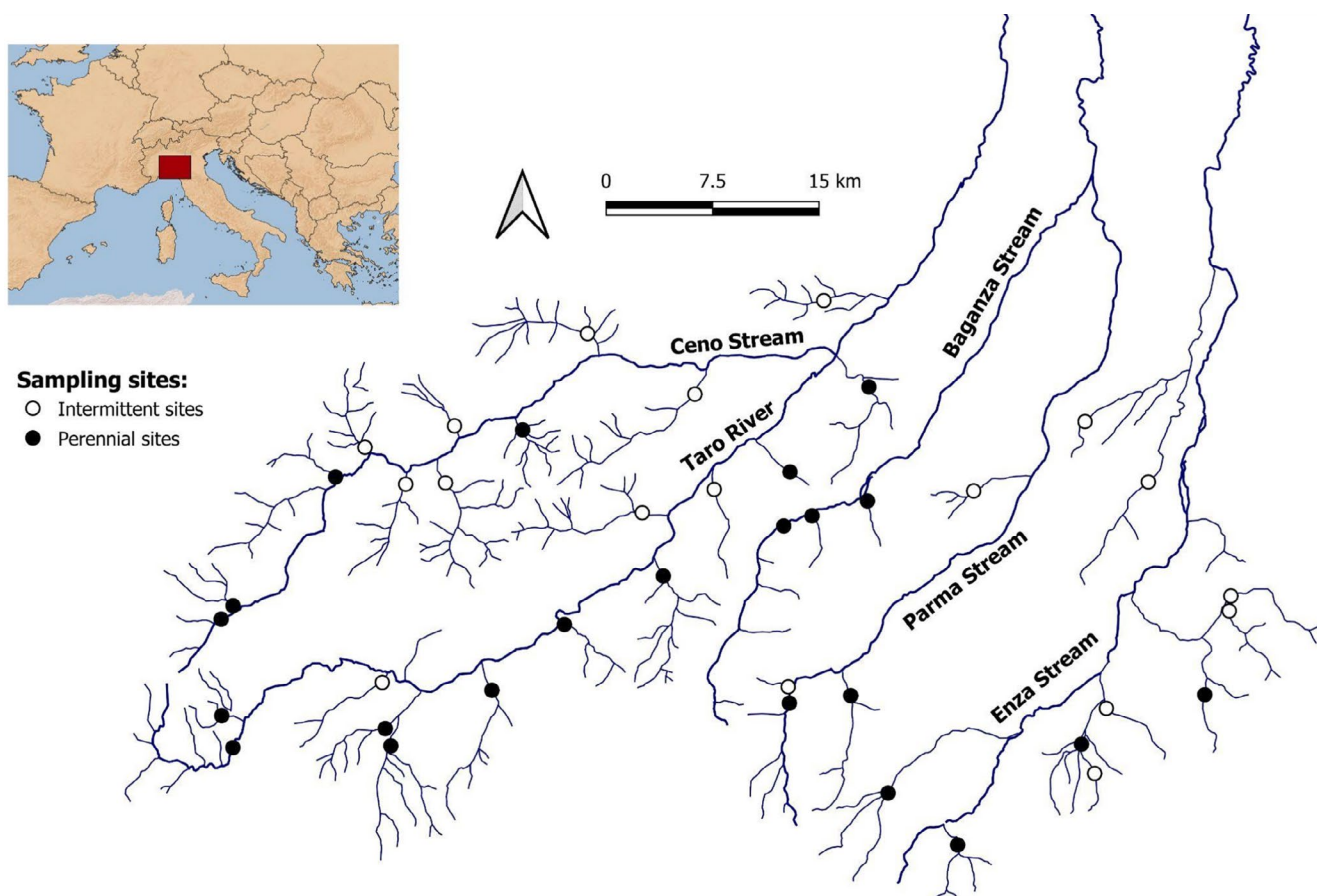


FIGURE 1 | Location of the 22 perennial and 18 intermittent sampling sites in the Po River basin, northern Italy. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/tra.4467)]

generalized LMM (GLMM, for richness metrics), the latter fitted with a Poisson distribution to account for zero-inflated count data (Zuur et al. 2009). *Flow regime* (categorical: P, I) was included as a fixed term, and *site* (categorical: 40 levels) as a random term. Model results were summarized with Type II ANOVA (to account for the unbalanced design), and *p*-values adjusted with Bonferroni correction for multiple comparisons.

3 | Results

We recorded 73,047 individuals in 133 taxa (87 families). Taxa with no DIS (30 taxa from 24 families) or MIS-group (52 taxa from 36 families) were excluded (Figure 2, Table S2), leaving 77% and 61% of taxa assigned to DIS and MIS-groups, respectively. Some excluded taxa occurred frequently, including the mayfly family Ephemerellidae (excluded from DEHLI calculations), present in 67% of the samples, and the mayfly genera *Ecdyonurus*, *Rhithrogena*, and *Habroleptoides* (excluded from MIS-index calculations), found in 64%, 39%, and 33% of the samples, respectively. Several excluded taxa had different occurrences in P and I-sites (Figure S1).

DEHLI index scores were significantly higher at P-sites (5.7 ± 0.7) than at I-sites (5.2 ± 0.8 , Table 1), whereas MIS-index values were comparable at both site types (Figure 3, Table 1). Differences in richness metrics between P and I-sites varied among DEHLI groups and MIS groups (Figure 3, Table 1).

Richness in the *flow-decrease* and *loss-longitudinal* DEHLI groups was significantly higher at P-sites (2 ± 1 and 5 ± 2 taxa per sample, respectively) than at I-sites (1 ± 1 and 3 ± 2 taxa per sample, respectively), whereas other DEHLI richness metrics were comparable across sites. The *lotic-fast* MIS group was significantly higher at P-sites (2 ± 1 taxa per sample) than at I-sites (1 ± 1 taxa per sample), whereas for all other MIS groups, richness was comparable at P and I-sites.

4 | Discussion and Conclusions

To date, the few indices developed to characterize how ecological communities respond to changing habitat availability during river drying remain largely untested beyond their region of origin. Testing is required to inform the adaptation and use of such indices, both to document responses to changing in-channel conditions and to inform the interpretation of bio-monitoring data (Stubbington et al. 2018). We thus evaluated the performance of two United Kingdom indices of intermittence (DEHLI, MIS-index) in Italy by comparing index scores and richness metrics representing macroinvertebrate communities from perennial (P) and intermittent (I) river sites. Differences in DEHLI scores and DEHLI and MIS-index richness metrics between P and I-sites in Italian rivers suggest that these indices have the potential to describe ecological responses to intermittence beyond their region of development. Metrics representing the richness of taxa associated with fast-flowing conditions were

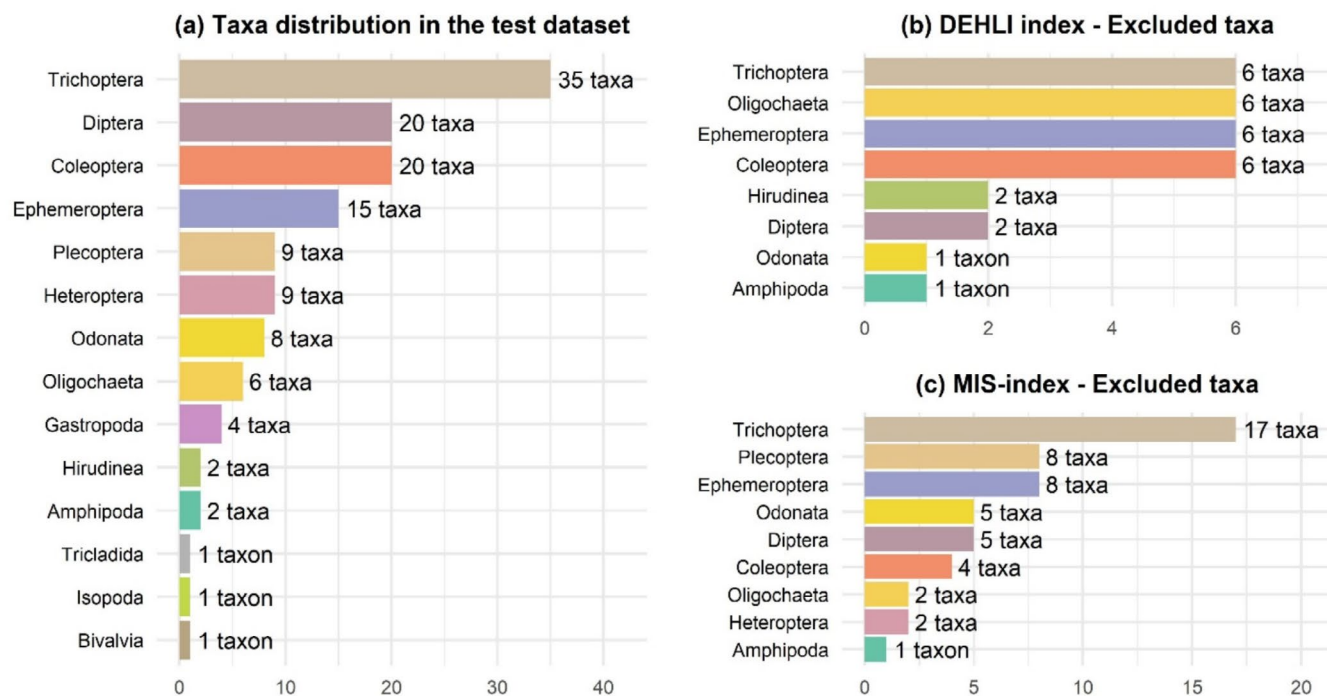


FIGURE 2 | Taxa distribution in the original test dataset (a), taxa without a DEHLI drought-intolerance score (b), and MIS-group (c), and therefore excluded from analyses. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/ra.4467)]

TABLE 1 | Results of mixed models summarized with type II ANOVA comparing index scores and richness metrics (which represent macroinvertebrate taxa richness in groups associated with different habitats) at perennial and intermittent sites.

	Chi-square	Df	p	Bonferroni adjusted p
DEHLI index and metrics				
DEHLI index	8.68	1	0.003	0.039
Flow-decrease	20.58	1	<0.001	<0.001
Loss-lateral	<0.01	1	0.997	1.000
Loss-longitudinal	15.33	1	<0.001	<0.001
Pool-shrinking	1.74	1	0.279	1.000
Pool-drying	0.23	1	0.635	1.000
MIS-index and metrics				
MIS-index	3.54	1	0.059	0.708
Lotic-fast	20.33	1	<0.001	<0.001
Lotic	3.63	1	0.057	0.681
Generalist	0.34	1	0.559	1.000
Lentic	2.55	1	0.110	1.000
Semi-aquatic	0.08	1	0.768	1.000

Note: Significant differences are marked in bold.

particularly responsive to intermittence, suggesting their potential use for assessing responses to drying events.

4.1 | Performance of Intermittence Indices

DEHLI index scores were significantly higher at P than I-sites, suggesting the potential of this index for application in regions beyond the United Kingdom. These differences between sites were detected based on macroinvertebrate communities sampled in late spring and early summer, approximately 7 months after the preceding dry phase and rewetting of the I-sites, similar to observations in United Kingdom calcareous rivers (Sarremejane et al. 2019; White et al. 2019). This suggests a “drying memory” of I-sites, with persistent effects on community composition, consistent with previous findings in temperate United Kingdom (Bunting et al. 2020), Mediterranean (Belmar et al. 2019), and Alpine (Piano et al. 2019) streams. Furthermore, we detected differences between P and I-sites for richness metrics representing groups associated with faster flow conditions, i.e., the *flow-decrease* and *loss-longitudinal* DEHLI groups and the *lotic-fast* MIS-group. Taxa in these groups prefer fast-flowing and well-oxygenated waters (Chadd et al. 2017; England et al. 2019) and thus may be eliminated during ponded and dry phases, with their recolonization after flow resumption reflecting taxon-specific dispersal capacities (Sarremejane et al. 2020). For example, many taxa belonging to the families Heptageniidae (Ephemeroptera), Leuctridae (Plecoptera), and Rhyacophilidae (Trichoptera), commonly found in both UK chalk streams and Italian rivers, prefer fast-flowing waters (e.g., Schmidt-Kloiber and Hering 2015) and were more abundant in P-sites, enhancing their potential as indicator taxa that represent the effects of stream drying. In contrast, the richness

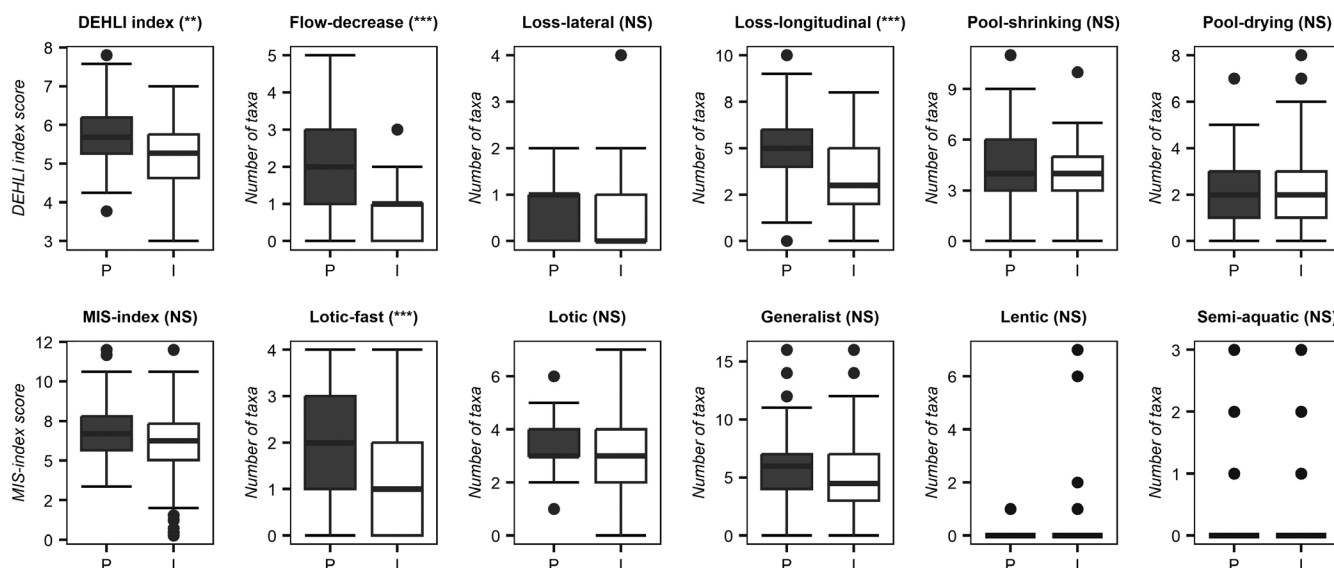


FIGURE 3 | DEHLI and MIS-index scores and richness metrics for habitat-association groups at perennial (P) and intermittent (I) sites. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; NS = non significant (i.e., $p > 0.05$).

of groups associated with later stages of drying, including lentic and semi-aquatic taxa (such as the Diptera family Tabanidae and the Heteroptera genus *Micronecta*), was comparable at P and I-sites. Differences in macroinvertebrate communities between P and I-sites thus mostly reflect the absence of rheophilic taxa from I-sites rather than the occurrence of lentic and semi-aquatic taxa in ponded and dry phases (Kelso and Entrekkin 2018).

4.2 | Factors Affecting Index Performance

The MIS-index and seven DEHLI and MIS-group richness metrics were comparable at P and I-sites. This difference in performance likely reflects their different taxonomic resolutions, with DEHLI's higher family level enhancing its transferability across different ecoregions (Heino and Soininen 2007). Similarly, the performance of the Lotic-invertebrate Index for Flow Evaluation (LIFE; Extence et al. 1999), developed for the United Kingdom, performs well in its family but not species-level form in other regions, including Mediterranean Greece (Theodoropoulos et al. 2021). Largely genus- and species-based indices such as the MIS-index have less taxonomic overlap among regions and are affected by local endemisms and biogeographic history. The resultant exclusion of 52 taxa from MIS-index calculations (39% of the dataset, compared to 18% for DEHLI) likely reduced its detection of differences between P and I-sites because several of the excluded taxa had different distributions between P and I-sites (Figure S2). In addition, several rheophiles including the Ephemeroptera genera *Rhithrogena*, *Oligoneuriella*, and *Ecdyonurus*, the Plecoptera genera *Perla*, *Dinocras*, and *Protonemura*, and the Diptera family Blephariceridae, as well as primarily lentic taxa including *Siphonurus* (Ephemeroptera), *Anax* and *Chalcolestes* (Odonata), and *Pomatinus* (Coleoptera), were present in the test dataset but are not MIS-index scoring taxa, potentially reducing the responsiveness of index scores and richness metrics to environmental differences between P and I-sites.

In addition, taxa in the *total-dry* DEHLI group and in the *semi-aquatic* and *terrestrial* MIS-groups were poorly represented in the Italian dataset, which was collected during flowing phases and exclusively within the wetted area (i.e., not also from marginal sediments, as recommended by England et al. 2019). Including semi-aquatic and terrestrial taxa may enhance characterization of ecological responses to flow intermittence (Bunting et al. 2020; Sarremejane et al. 2019) but can also pose challenges, requiring specialized field and taxonomic identification skills (England et al. 2019).

The performance of the DEHLI and MIS indices (and derived richness metrics) may also reflect contrasting characteristics of the drying process in different river morphologies. Both indices were developed and tested using data from chalk streams in south and east England (Chadd et al. 2017; England et al. 2019), which are groundwater-dominated, narrow, single-channel rivers (Berrie 1992). In contrast, we sampled Italian rivers fed both by surface runoff and groundwater, and typified by wider, multi-channel. The wetted part of the riverbed is highly mobile and lacks vegetated margins in these braided rivers (Vander Vorste et al. 2016). As a result, the DEHLI stage *loss-lateral*—in which lateral connectivity is lost in chalk streams, eliminating an associated group of taxa that use marginal habitats, including their vegetation—does not occur consistently in braided intermittent rivers. As such, taxa included in this DEHLI group were scarce in our dataset, likely preventing its discrimination of P and I-sites.

4.3 | Future Perspectives

Our study highlights opportunities to adapt the tested indices to improve their performance and thus support their future application in different ecoregions. For instance, index performance could be enhanced (i) by regional adaptation of the taxa list to better represent taxa pools, including scoring and classification of new taxa, (ii) by considering differences in river morphology

and drying patterns. Moreover, the dynamic mosaic of aquatic and terrestrial habitats that characterizes intermittent rivers represents an opportunity to include more semi-aquatic and river-associated terrestrial taxa in intermittence indices.

Considering the current and predicted increase in river drying (Mimeau, Künne, Devers et al. 2024b; Zipper et al. 2021), indices are needed to document the ecological effects of drying on stream biota, ideally, indices that remain effective across regions and river types. To overcome regional differences in taxa pool and thus enhance transferability, indices calculated based on selected functional traits represent a promising solution, as demonstrated by Laini et al. (2022) using their Flow-T index. Like taxonomic indices, such trait-based metrics require testing beyond their region of origin, to determine their wider applicability, limits and potential for improvement or regional adaptation. In addition, such indices could inform the interpretation of biomonitoring indices that document biotic responses to human pressures, thus supporting ecosystem health assessment in intermittent rivers.

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.